Imperial College London

21 October 2026

Drift to the Posterior

One-Step Simulation-Based Inference with Drifting Models

Michele Cespa Charlotte Heard Borna Khodabandeh Sean White

Imperial College London

Outline

1. Motivation: Bayesian inference without a likelihood
2. Flow Matching
3. Drifting Models
4. Experiments
5. Conclusions

Imperial College London

The inference problem

- Goal: the posterior

$$p(\theta \mid x) = \frac{p(x \mid \theta)\, p(\theta)}{p(x)}$$

- MCMC / VI need the unnormalised density $p(x \mid \theta)\, p(\theta)$
- Often $p(x \mid \theta)$ is only defined *implicitly* by a simulator
- **Simulation-based inference (SBI):** learn $q(\theta \mid x)$ from simulated pairs (θ, x)

Posterior estimation as generative modelling

- If we can *sample* $\theta \sim q(\cdot \mid x)$ for any x, we have an amortised posterior
- Generative modelling recipe:
  1. Sample noise $\varepsilon \sim \mathcal{N}(0, I)$
  2. Push through a learned map f
  3. Train f so that $f_{\#}\, p_0 = p_{\text{data}}$
- Two ways to build f: **Flow Matching** (many steps) vs **Drifting** (one step)

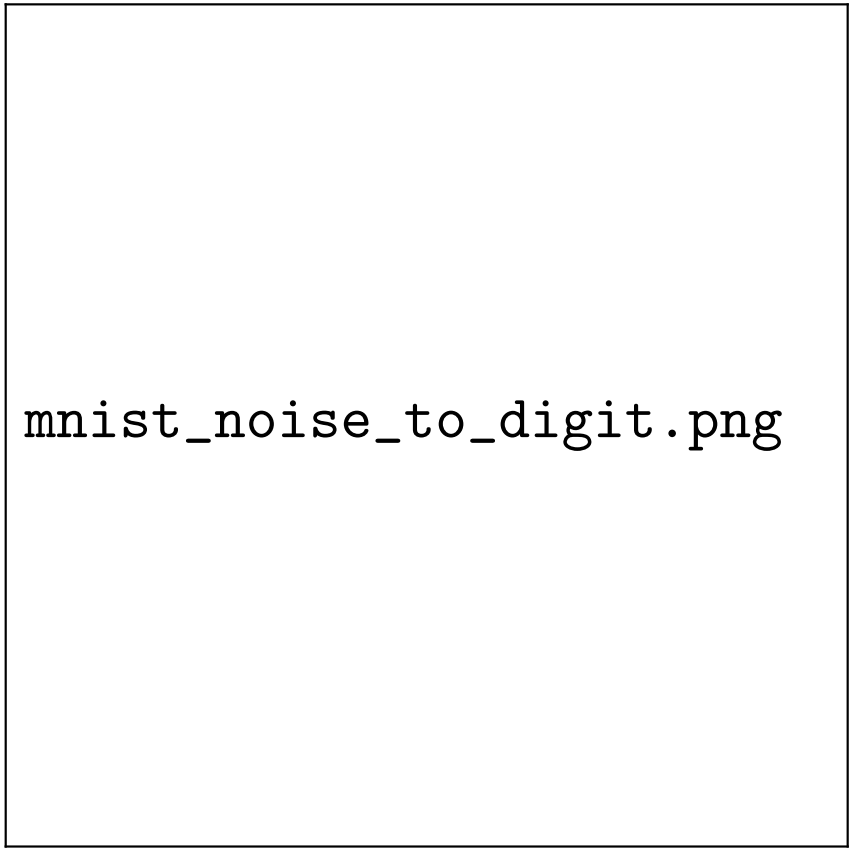
Imperial College London

Flow matching: a generative model (MNIST)

- Data: images $x_1 \sim p_{\text{data}}$ (MNIST digits)
- Want more samples from p_{data}
- Start from noise $x_0 \sim \mathcal{N}(0, I_d)$
- Move it along an ODE

$$\frac{dx_t}{dt} = u_t(x_t)$$

so that $x_1 \sim p_{\text{data}}$



Imperial College London

Continuous flows

- A probability path p_t, $t \in [0,1]$: $p_0 = \mathcal{N}(0, I)$, $p_1 \approx p_{\text{data}}$
- Velocity field u_t and density linked by the continuity equation

$$\partial_t p_t + \nabla \cdot (p_t u_t) = 0$$

- Learn $v_t^\phi \approx u_t$ with a neural network:

$$\mathcal{L}_{\text{FM}}(\phi) = \mathbb{E}_{t,\, x_t \sim p_t} \left\| v_t^\phi(x_t) - u_t(x_t) \right\|^2$$

- Problem: u_t and p_t are unknown

The key trick: condition on one data point

- Choose a simple path per data point x_1:

$$p_t(x \mid x_1) = \mathcal{N}\big(x \mid t x_1,\ \sigma_t^2 I\big), \qquad \sigma_t = 1 - (1 - \sigma_{\min})t$$

- Its vector field is known in closed form (straight line, OT path):

$$u_t(x \mid x_1) = \frac{x_1 - (1 - \sigma_{\min})x}{\sigma_t}$$

- Conditional flow matching loss has the **same minimiser** as $\mathcal{L}_{\text{FM}}$:

$$\mathcal{L}_{\text{CFM}}(\phi) = \mathbb{E}_{t,\, x_1,\, x_t \sim p_t(\cdot \mid x_1)} \left\| v_t^{\phi}(x_t) - u_t(x_t \mid x_1) \right\|^2$$

Imperial College London

Training and sampling

Training step

1. $x_1 \sim p_{\text{data}}$ (an image)
2. $t \sim \mathcal{U}[0,1], \;\; x_0 \sim \mathcal{N}(0, I)$
3. $x_t = \sigma_t x_0 + t x_1$
4. Regress $v_t^{\phi}(x_t)$ onto $x_1 - (1 - \sigma_{\min})x_0$

Sampling

1. $x_0 \sim \mathcal{N}(0, I)$
2. Solve $\dot{x}_t = v_t^{\phi}(x_t)$ from $t = 0$ to 1
3. Many network evaluations per sample

From images to posteriors

- Generative modelling: we *have* samples from the target p_{data}
- SBI: target is $p(\theta \mid x)$ — **no posterior samples**
- Bayes: $p(x)\, p(\theta \mid x) = p(\theta)\, p(x \mid \theta)$, so

$$\mathbb{E}_{p(x)\, p(\theta|x)} \longrightarrow \mathbb{E}_{p(\theta)\, p(x|\theta)}$$

- Simulate pairs: $\theta \sim p(\theta)$, then $x \sim p(x \mid \theta)$

Change of roles

image $x_1 \rightarrow$ parameters θ_1, and the network is *conditioned* on the observation x

Imperial College London

Flow Matching Posterior Estimation (FMPE)

- Loss (Dax et al., 2023):

$$\mathcal{L}_{\text{FMPE}} = \mathbb{E}_{t,\,\theta_1,\,x,\,\theta_t} \left\| v_{t,x}(\theta_t) - u_t(\theta_t \mid \theta_1) \right\|^2$$

  with $\theta_1 \sim p(\theta)$, $x \sim p(x \mid \theta_1)$, $\theta_t \sim p_t(\theta_t \mid \theta_1)$
- Network $v_{t,x}(\theta_t)$ takes (t, θ_t, x) and outputs a vector of dimension $\dim \theta$
- Amortised: train once, sample $q(\theta \mid x_o)$ for any new x_o
- Exact density via $\log q = \log p_0 - \int_0^1 \nabla \cdot v_{t,x}\, dt$
- Mass-covering: $D_{\text{KL}}(p \,\|\, q) \leq C\, \text{MSE}^{1/2}$

Imperial College London

Drifting: a one-step generator

- Learn f_ϕ directly: $x = f_\phi(\varepsilon)$ — **one forward pass**
- Drifting field: attract to data (positives), repel from model samples (negatives)

$$V_{p,q}(x) = V_p^+(x) - V_q^-(x)$$

- Stop-gradient loss:
  $\mathcal{L}(\phi) = \mathbb{E}_\varepsilon \left\| f_\phi(\varepsilon) - \mathrm{sg}\big(f_\phi(\varepsilon) + V_{p,q}(f_\phi(\varepsilon))\big) \right\|^2$

Imperial College London

Joint Drifting for SBI (ours)

- Problem: the drifting field needs many samples from $p(\theta \mid x)$ — we only have one θ per x
- Solution: drift on the **joint** space (θ, x)
- Product kernel $k_\Theta(\theta, \theta')\, k_X(x, x')$; only the θ-block enters the loss
- Prior draws $\theta_0 \sim p(\theta)$ as the base distribution

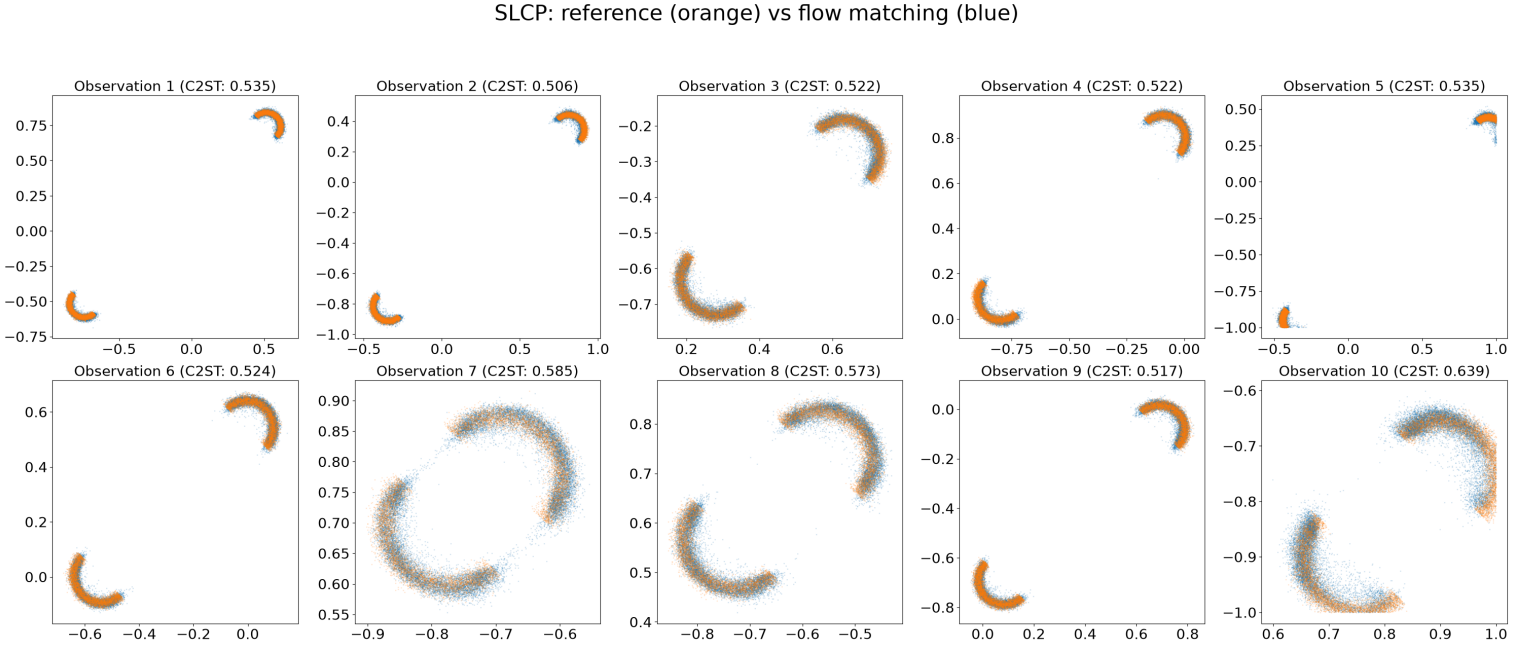
Imperial College London

Setup

- Benchmark: `sbibm` — Two Moons, SLCP, Lotka–Volterra
- Metric: C2ST (0.5 = indistinguishable from reference, 1 = fully separable)
- FMPE: residual MLP, $\sim$8.9M parameters, $N = 10^5$ simulations
- Drifting: FiLM residual MLP, $\sim$1.2M parameters, fresh simulations each step

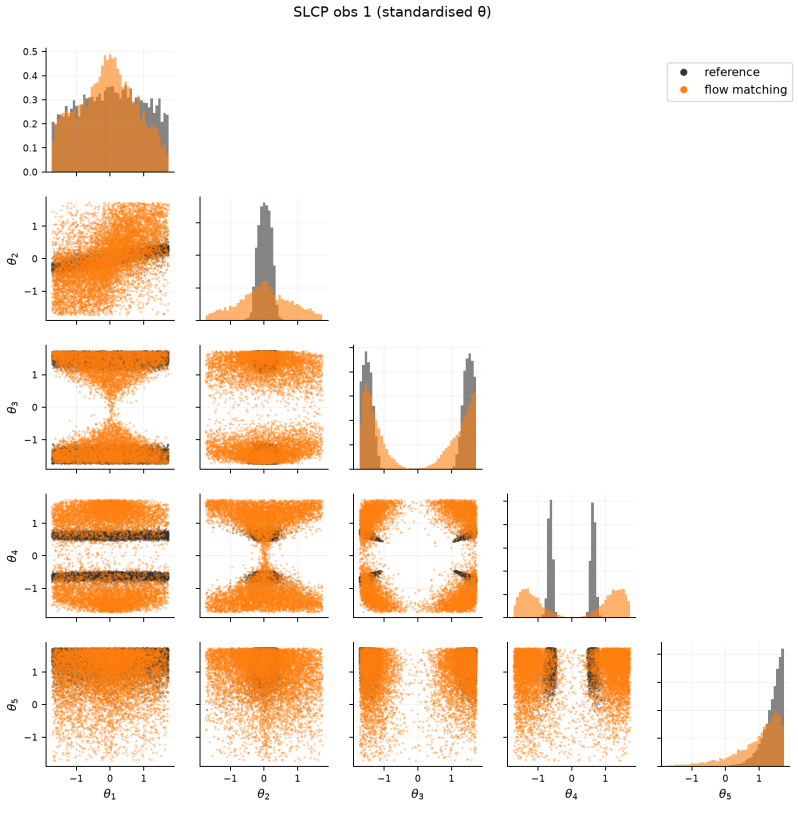
Imperial College London

Two Moons



Imperial College London

SLCP



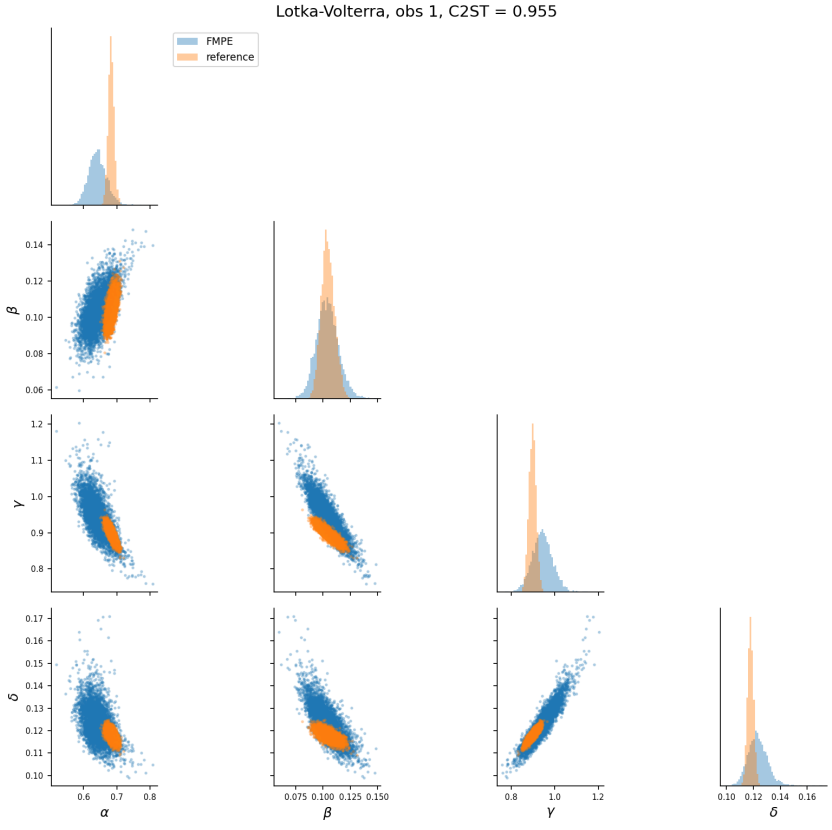
Flow matching

drifting_SLCP_marginals.png

Drifting

Imperial College London

Lotka–Volterra



FMPE, observation 1

drifting_lv_predictive.png

Drifting, posterior predictive

Imperial College London

Results summary

Task	FMPE C2ST	Drifting C2ST
Two Moons	–	–
SLCP	–	–
Lotka–Volterra	–	–
Network passes per sample	$\sim$250	1

Imperial College London

Conclusions

- Flow matching: accurate, mass-covering, exact density — but many-step sampling
- Drifting: one-step sampling, fast posterior predictive checks
- Joint Drifting adapts drifting to SBI without conditional samples
- Next steps: same test observations and references for both methods, calibration (SBC)

Thank you — questions?